

Thematic Assessment Report
Phase 3

TA REPORT PHASE 3

Persistent Memory Architecture in Agents Using DBMS

Implementation (Python · SQLite · FTS5)

Submitted in partial fulfilment of course requirements
for Database Management Systems

Bachelor of Technology

Computer Science and Engineering
School of Engineering and Technology



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Declaration. We declare that this report and the program in `ta3-implementation/` are our own work for DBMS TA-3.

1. Introduction and objective

Phase 1 of this term paper reviewed persistent memory for AI agents. Phase 2 compared Notion's Lore with the same logical schema on SQLite and PostgreSQL, and added Nemori, LightMem, and *Harness the Memory*. The recommendation was a local relational store: episodic rows, semantic triples with a validity interval, a normalised entity and alias table, provenance back to the episode that supported a claim, and indexes for keyword and vector retrieval. Lore's vault stays readable, but it cannot declare those constraints. This phase implements that store.

The objective is a small program a reader can run. It has to support write, retrieve (keyword and vector), consolidate, update, and forget, on a schema with keys and checks, and print the result of one agent session.

2. Software and tools

The memory store imports only the Python standard library. The versions below were read on the machine that produced the screenshots.

Software	Version	Role
Python	3.12.3	<code>sqlite3</code> is in the standard library, so <code>python3 demo.py</code> needs no extra packages.
SQLite	3.45.1	One file, with foreign keys, checks, a partial unique index, and FTS5. No server.
sqlite3 CLI	3.45.1	Runs <code>queries.sql</code> at the end of the demo.
VS Code	1.139.1	Editor used for the code screenshots.
xfce4-terminal	xfce4-terminal 1.1.3 (Xfce 4.18)	Terminal used to run <code>python3 demo.py</code> .
Graphviz	2.43.0 (0)	<code>dot</code> and <code>neato</code> drew the diagrams.
ReportLab	5.0.1	Prints this PDF. Not used by the demo.
Linux	Ubuntu 24.04.4 LTS; kernel 6.12.94+	The machine the demo was run on (x86_64).

PostgreSQL would add a temporal exclusion constraint, which SQLite cannot express. That is noted under future work. SQLite is enough to show primary keys, foreign keys, checks, and a partial unique index, and the program runs from a clone with no extra install.

3. System architecture

An agent session does not talk to SQL itself. `demo.py` calls `MemoryStore`. The class opens one SQLite connection, turns foreign keys on (SQLite leaves them off unless asked), and applies `schema.sql`. Each operation is a short transaction. Retrieval is three queries whose ranks are fused in Python with reciprocal rank fusion, the same rule Phase 2 noted in Lore, with the constant $k = 60$.

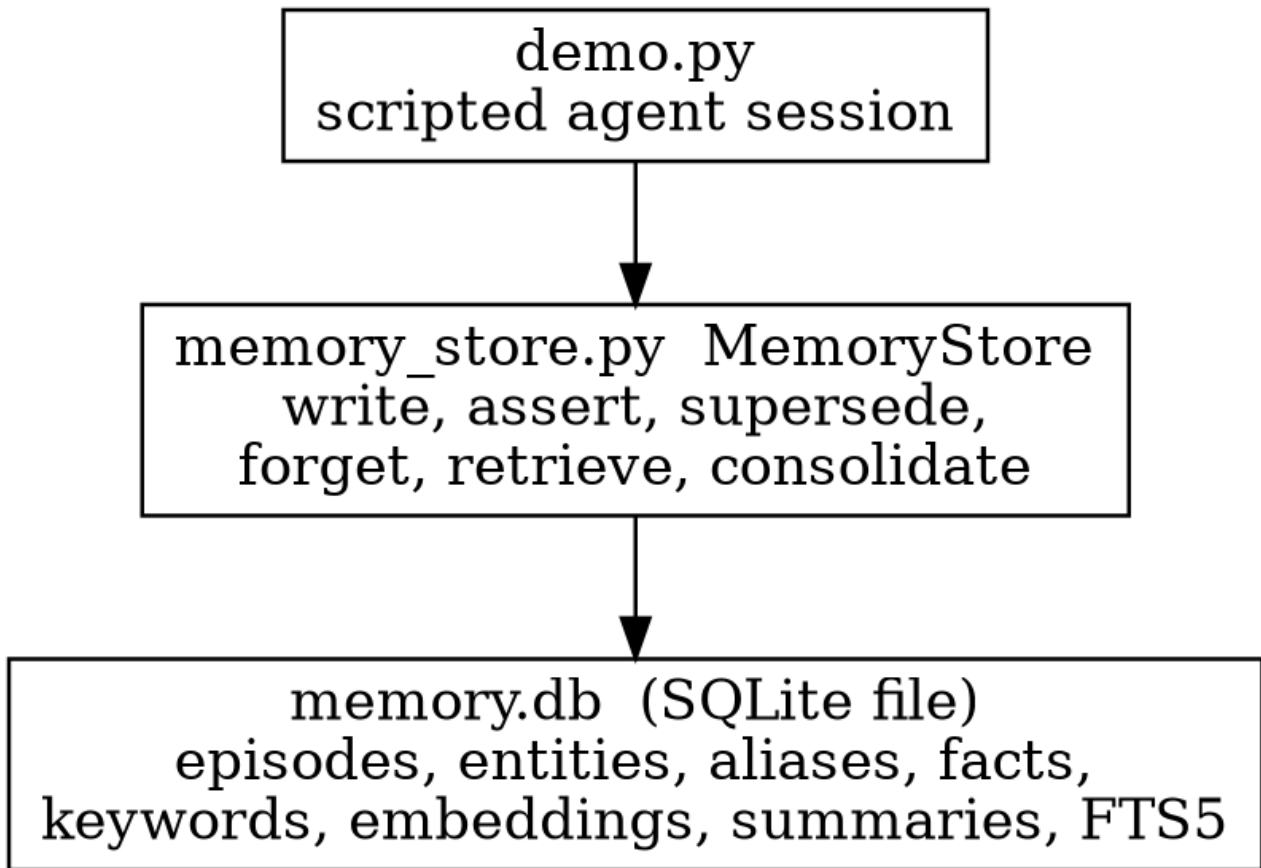


Figure 1: How the pieces fit. `demo.py` calls `MemoryStore`, and that module is the only writer of `memory.db`.

The six memory operations from the Phase 2 survey map onto the schema as follows. Write inserts an episode and indexes it. Assert inserts a semantic fact. Supersede is the update: the old row's interval is closed and a new row is inserted. Forget changes a status flag and keeps the row, so provenance still joins. Retrieve reads. Consolidate inserts a summary row. That summary is the active facts from the session, not a second copy of the episodes.

4. Database schema

Figure 2 is the ER diagram of the memory store, drawn in Chen notation. An entity is a rectangle, an attribute is an ellipse, and a relationship is a diamond. Primary keys are underlined. Cardinality (1 or N) is written on the lines. `alias` and `keyword` are multi-valued, so they are double ellipses. Embedding depends on the episode, fact, or summary it describes, so it is a weak entity (double rectangle) and `{code('embeds')}` is a double diamond. A later fact that replaces an earlier one is the `{code('succeeds')}` relationship. In the tables, a multi-valued attribute is its own rows: `alias_key` is a primary key, and `keyword` has a check that exactly one parent is set.

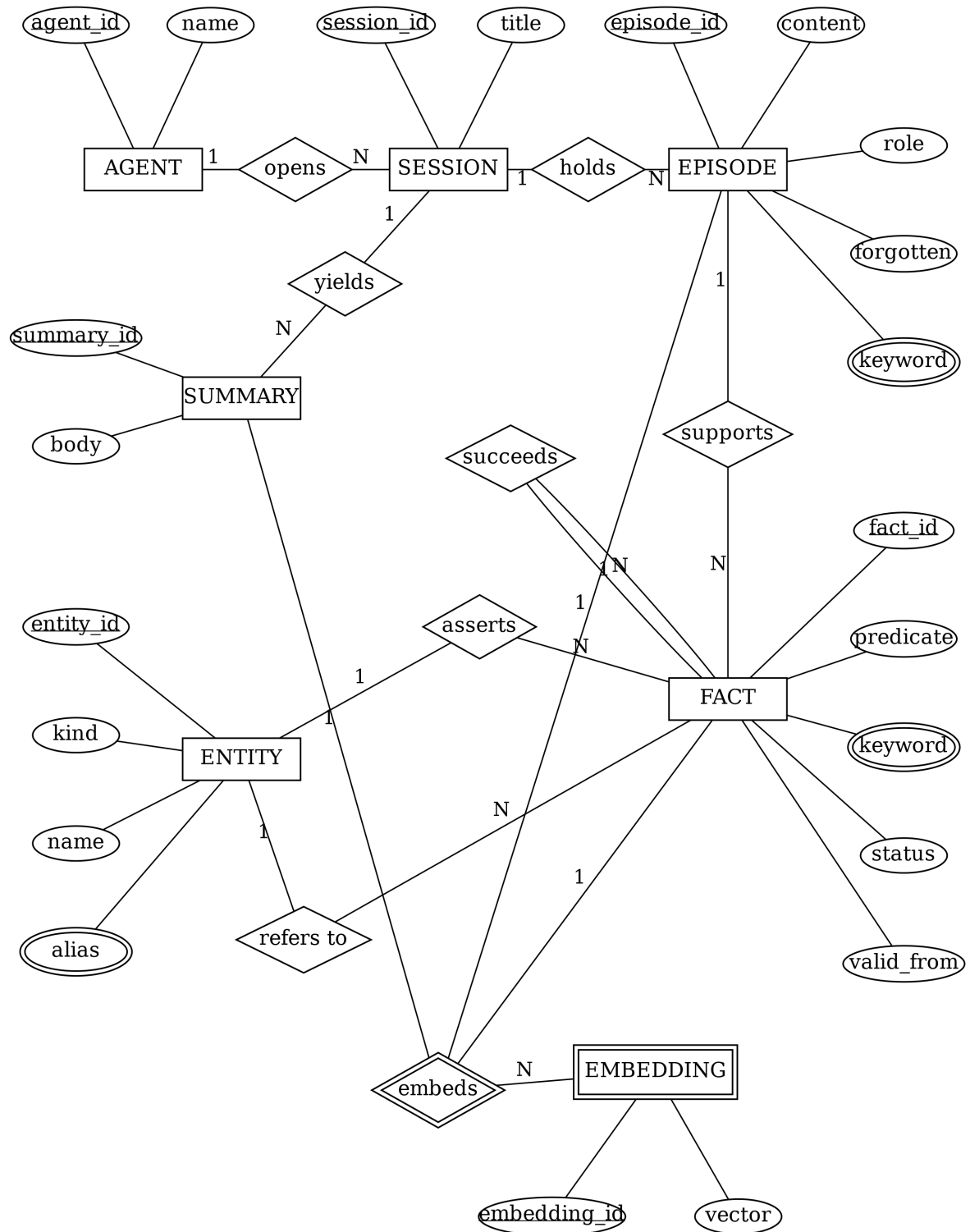


Figure 2: ER diagram of the memory store.

FACT is the semantic memory. **valid_from** and **valid_until** are valid time. A check requires an active fact to have an open interval, and a superseded or forgotten fact to have a closed one. The partial unique index `ux_one_active_functional_fact` allows many historical owners and at most one active `owned_by`, `is_a`, or `created_by` per subject. `uses` is not in that index: an application may use

more than one system, and a wrong guess is forgotten rather than blocked. Views `belief_history` and `current_belief` are what the as-of query and the current-owner query read.

```
61 CREATE TABLE fact (  
62     fact_id          INTEGER PRIMARY KEY,  
63     subject_entity_id INTEGER NOT NULL REFERENCES entity(entity_id),  
64     predicate        TEXT NOT NULL  
65     CHECK (predicate IN (  
66         'is_a', 'uses', 'owned_by', 'created_by', 'depends_on'  
67     )),  
68     object_text      TEXT,  
69     object_entity_id INTEGER REFERENCES entity(entity_id),  
70     confidence        REAL NOT NULL CHECK (confidence > 0 AND confidence <= 1),  
71     stance           TEXT NOT NULL  
72     CHECK (stance IN ('certain', 'likely', 'speculative')),  
73     valid_from       TEXT NOT NULL,  
74     valid_until      TEXT,  
75     status           TEXT NOT NULL DEFAULT 'active'  
76     CHECK (status IN ('active', 'superseded', 'forgotten')),  
77     source_episode_id INTEGER REFERENCES episode(episode_id),  
78     superseded_by    INTEGER REFERENCES fact(fact_id),  
79     statement        TEXT NOT NULL,  
80     created_at       TEXT NOT NULL,  
81     CHECK (object_text IS NOT NULL OR object_entity_id IS NOT NULL),  
82     CHECK (valid_until IS NULL OR valid_from <= valid_until),  
83     CHECK (  
84         (status = 'active' AND valid_until IS NULL)  
85         OR (status IN ('superseded', 'forgotten') AND valid_until IS NOT NULL)  
86     )  
87 );  
88  
89 -- At most one *current* object for predicates that are functions of the  
90 -- subject. This is the constraint a conflict scan otherwise has to rediscover.  
91 CREATE UNIQUE INDEX ux_one_active_functional_fact  
92     ON fact (subject_entity_id, predicate)  
93     WHERE status = 'active'  
94     AND predicate IN ('owned_by', 'is_a', 'created_by');  
95  
96 CREATE INDEX ix_fact_source ON fact (source_episode_id);  
97 CREATE INDEX ix_fact_validity ON fact (valid_from, valid_until);  
98 CREATE INDEX ix_fact_object ON fact (object_entity_id);
```

Figure 3: The `fact` table and the partial unique index in `schema.sql`, open in VS Code. An active fact must have an open interval. The index allows many past owners and only one current owned_by, is_a, or created_by.

5. Implementation

`memory_store.py` is the only module. Tokens are lower-cased, stopwords are dropped, and a short suffix stem (owns and owned both become own) makes the keyword table agree with the FTS5 porter tokenizer. Embeddings are a signed feature hash into 512 buckets, using SHA-256 so the vector does not change between processes. Cosine similarity is the dot product of two unit vectors. There is no neural model and no network call. The embedding table stores a JSON array of floats, which `sqlite3` can print.

`assert_fact` resolves names through `entity_alias`, so the session can say "the app" and hit Campus Navigator. If a functional predicate already has a different active object, the method raises `FactConflict` and does not insert. `supersede_fact` then closes the old interval and inserts the replacement in one transaction: it marks the old row superseded, inserts the new row, and sets

superseded_by. The unique index is never asked to hold two active owners. Figure 8 shows what a raw second insert does.

```

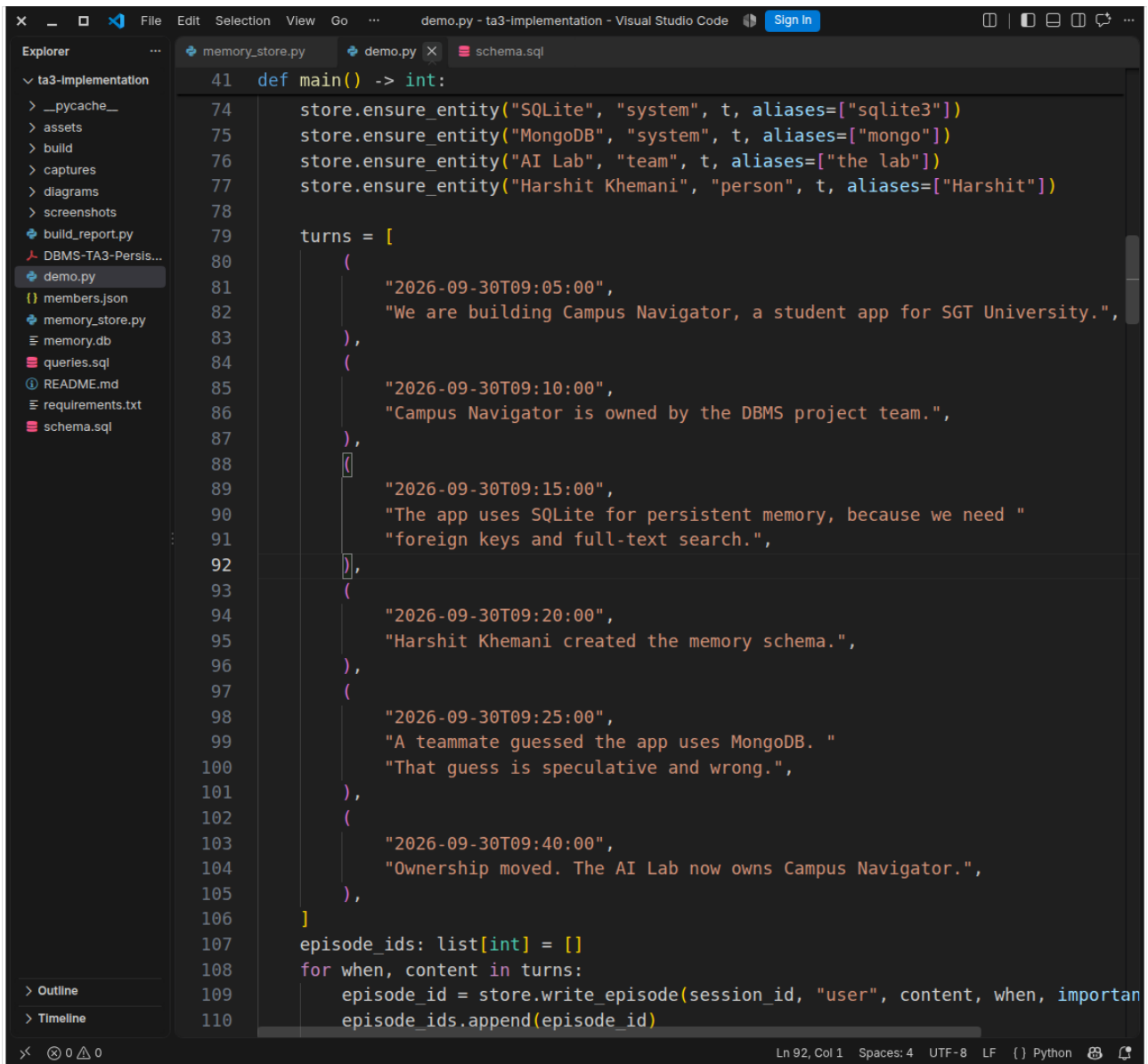
105 class MemoryStore:
359     def consolidate(self, session_id: int, when: str) -> int:
403         self._store_embedding(embed(body), summary_id=summary_id)
404         self.conn.commit()
405         return summary_id
406
407     def retrieve(self, query: str, limit: int = 5) -> list[dict[str, object]]:
408         tokens = tokenize(query)
409         lanes = {
410             "fts": self._fts_keys(tokens),
411             "keyword": self._keyword_keys(tokens),
412             "vector": self._vector_keys(query),
413         }
414         scores: dict[str, float] = {}
415         used: dict[str, list[str]] = {}
416         for lane, keys in lanes.items():
417             for rank, key in enumerate(keys, start=1):
418                 scores[key] = scores.get(key, 0.0) + 1.0 / (RRF_K + rank)
419                 used.setdefault(key, []).append(lane)
420         ordered = sorted(scores, key=lambda key: (-scores[key], key))
421         hits: list[dict[str, object]] = []
422         for key in ordered[:limit]:
423             kind, raw_id = key.split(":", 1)
424             hits.append(
425                 {
426                     "kind": kind,
427                     "id": int(raw_id),
428                     "score": scores[key],
429                     "lanes": "+".join(used[key]),
430                     "text": self._label(kind, int(raw_id)),
431                 }
432             )
433         return hits
434
435     def current_beliefs(self) -> list[sqlite3.Row]:
436         return list(
437             self.conn.execute(
438                 """

```

Figure 4: retrieve in memory_store.py. Three ranked lists are fused with reciprocal rank fusion, $k = 60$. The lanes field is what the transcript prints.

forget_fact sets status to forgotten and writes valid_until. forget_episode sets forgotten and deletes that episode's keyword rows. The episode row stays, because a fact may still cite it. Deleting the episode outright is refused by the foreign key, which the demo provokes and then rolls back. consolidate inserts one summary row whose text is the active facts sourced from the session, with their confidences. That is a stored row, not a language-model summary.

demo.py is the session. The clock is a list of fixed timestamps, so a second run prints the same transcript. Figure 5 is the part that registers entities and the six turns.



```
41 def main() -> int:
74     store.ensure_entity("SQLite", "system", t, aliases=["sqlite3"])
75     store.ensure_entity("MongoDB", "system", t, aliases=["mongo"])
76     store.ensure_entity("AI Lab", "team", t, aliases=["the lab"])
77     store.ensure_entity("Harshit Khemani", "person", t, aliases=["Harshit"])
78
79     turns = [
80         (
81             "2026-09-30T09:05:00",
82             "We are building Campus Navigator, a student app for SGT University.",
83         ),
84         (
85             "2026-09-30T09:10:00",
86             "Campus Navigator is owned by the DBMS project team.",
87         ),
88         (
89             "2026-09-30T09:15:00",
90             "The app uses SQLite for persistent memory, because we need "
91             "foreign keys and full-text search.",
92         ),
93         (
94             "2026-09-30T09:20:00",
95             "Harshit Khemani created the memory schema.",
96         ),
97         (
98             "2026-09-30T09:25:00",
99             "A teammate guessed the app uses MongoDB. "
100             "That guess is speculative and wrong.",
101         ),
102         (
103             "2026-09-30T09:40:00",
104             "Ownership moved. The AI Lab now owns Campus Navigator.",
105         ),
106     ]
107     episode_ids: list[int] = []
108     for when, content in turns:
109         episode_id = store.write_episode(session_id, "user", content, when, important)
110         episode_ids.append(episode_id)
```

Figure 5: The scripted session in `demo.py`. Aliases are registered with the entity, and each turn is one episode.

6. Output and results

`python3 demo.py` deletes any existing `memory.db`, applies the schema, and runs the session. The figures in this section are screenshots of that run in `xfce4-terminal`. The same transcript is saved in `captures/demo-stdout.txt` when this PDF is rebuilt.

```
ta3-demo
File Edit View Terminal Tabs Help
ta3-implementation $ python3 demo.py

=====
1. Schema
=====
database file: memory.db
tables: agent, embedding, entity, entity_alias, episode, episode_fts,
episode_fts_config, episode_fts_data, episode_fts_docsize, episode_fts_idx, fact,
fact_fts, fact_fts_config, fact_fts_data, fact_fts_docsize, fact_fts_idx, keyword,
session, summary
partial unique index ux_one_active_functional_fact:
CREATE UNIQUE INDEX ux_one_active_functional_fact ON fact (subject_entity_id, predicate)
WHERE status = 'active' AND predicate IN ('owned_by', 'is_a', 'created_by')

=====
2. Write a session
=====
agent 1 Archivist, session 1
episode 1 2026-09-30T09:05:00 We are building Campus Navigator, a student app for SGT
University.
episode 2 2026-09-30T09:10:00 Campus Navigator is owned by the DBMS project team.
episode 3 2026-09-30T09:15:00 The app uses SQLite for persistent memory, because we
need foreign keys and full-text search.
episode 4 2026-09-30T09:20:00 Harshit Khemani created the memory schema.
episode 5 2026-09-30T09:25:00 A teammate guessed the app uses MongoDB. That guess is
speculative and wrong.
episode 6 2026-09-30T09:40:00 Ownership moved. The AI Lab now owns Campus Navigator.
alias 'the app' -> entity 1 Campus Navigator (project)

=====
```

Figure 6: The start of python3 demo.py. The partial unique index is printed from sqlite_master, then the six episodes. The alias the app resolves to entity 1, Campus Navigator.

```
ta3-demo
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need foreign keys and full-text search.
episode 4 2026-09-30T09:20:00 Harshit Khemani created the memory schema.
episode 5 2026-09-30T09:25:00 A teammate guessed the app uses MongoDB. That guess is
speculative and wrong.
episode 6 2026-09-30T09:40:00 Ownership moved. The AI Lab now owns Campus Navigator.
alias 'the app' -> entity 1 Campus Navigator (project)

=====
3. Assert semantic facts
=====
fact 1 Campus Navigator owned_by DBMS project team certain 0.90 from
2026-09-30T09:10:00
fact 2 Campus Navigator uses SQLite certain 0.95 from 2026-09-30T09:15:00
fact 3 Campus Navigator created_by Harshit Khemani certain 0.90 from
2026-09-30T09:20:00
fact 4 Campus Navigator uses MongoDB speculative 0.30 from 2026-09-30T09:25:00

=====
4. Update and forget
=====
API refused a second active owner (fact 1 is still current)
existing: Campus Navigator owned by DBMS project team
supersede fact 1 -> fact 5 at 09:40
forget fact 4 (speculative MongoDB use) at 09:42
forget episode 5 (the wrong guess is kept in the table, hidden from retrieval)
ownership history:
fact 1 DBMS project team superseded [2026-09-30T09:10:00 .. 2026-09-30T09:40:00]
fact 5 AI Lab active [2026-09-30T09:40:00 .. open]

=====
```

Figure 7: Facts, then the update. The API refuses a second active owner. Supersede closes fact 1 at 09:40 and inserts fact 5. Fact 4 and episode 5 (the MongoDB guess) are forgotten. The history shows fact 1 superseded and fact 5 active.

```
ta3-demo
File Edit View Terminal Tabs Help

5. The index refuses what the API already refused
=====
IntegrityError: UNIQUE constraint failed: fact.subject_entity_id, fact.predicate
=====

6. Retrieve
=====
predicate lookup owned_by(Campus Navigator) = AI Lab (fact 5, from 2026-09-30T09:40:00)

query: full-text search sqlite
  1. episode id=3 rrf=0.0492 lanes=fts+keyword+vector
     user: The app uses SQLite for persistent memory, because we need foreign keys and
         full-text search.
  2. fact id=2 rrf=0.0484 lanes=fts+keyword+vector
     Campus Navigator uses SQLite

query: who owns Campus Navigator
  1. episode id=2 rrf=0.0489 lanes=fts+keyword+vector
     user: Campus Navigator is owned by the DBMS project team.
  2. episode id=6 rrf=0.0481 lanes=fts+keyword+vector
     user: Ownership moved. The AI Lab now owns Campus Navigator.
  3. fact id=5 rrf=0.0479 lanes=fts+keyword+vector
     Campus Navigator owned by AI Lab
  4. episode id=1 rrf=0.0469 lanes=fts+keyword+vector
     user: We are building Campus Navigator, a student app for SGT University.
  5. fact id=2 rrf=0.0464 lanes=fts+keyword+vector
     Campus Navigator uses SQLite

query: MongoDB
(no hit)
```

Figure 8: A raw INSERT of another active owned_by fails with UNIQUE constraint failed. The predicate query returns AI Lab, fact 5. Lexical fusion ranks the old ownership episode first and the current fact third. After the MongoDB rows are forgotten, the query MongoDB returns no hit.

```
ta3-demo
File Edit View Terminal Tabs Help

=====
7. Consolidate
=====
summary 1 over 5 live episodes
Consolidated beliefs from this session: Campus Navigator uses SQLite (confidence 0.95);
Campus Navigator created by Harshit Khemani (confidence 0.90); Campus Navigator owned by
AI Lab (confidence 0.92).

=====
8. Provenance delete is refused
=====
DELETE episode 3 refused: FOREIGN KEY constraint failed

=====
9. Checks
=====
PASS alias 'the app' is Campus Navigator
PASS exactly one active owner
PASS as-of 09:30 the owner is the DBMS project team
PASS MongoDB fact is forgotten and not a current belief
PASS retrieval of 'full-text search sqlite' hits a SQLite memory
PASS forgotten MongoDB memory is not retrieved
PASS supersede wrote a new fact id
PASS summary names the AI Lab
PASS PRAGMA integrity_check is ok

=====
10. sqlite3 queries
=====
```

Figure 9: Consolidation, the foreign-key refusal, and the nine checks. The summary names SQLite, Harshit Khemani, and AI Lab, and does not name MongoDB. Deleting episode 3, which fact 2 cites, is refused.

```

== tables ==
agent          episode_fts    fact_fts_config
belief_history  episode_fts_config fact_fts_data
current_belief episode_fts_data fact_fts_docsize
embedding      episode_fts_docsize fact_fts_idx
entity         episode_fts_idx keyword
entity_alias   fact          session
episode        fact_fts      summary

== current beliefs ==
fact_id | subject | predicate | object | stance | confidence | valid_from
2 | Campus Navigator | uses | SQLite | certain | 0.95 | 2026-09-30T09:15:00
3 | Campus Navigator | created_by | Harshit Khemani | certain | 0.90 | 2026-09-30T09:20:00
5 | Campus Navigator | owned_by | AI Lab | certain | 0.92 | 2026-09-30T09:40:00

== who owned Campus Navigator at 09:30, before the hand-off ==
subject | object | status | valid_from | valid_until
Campus Navigator | DBMS project team | superseded | 2026-09-30T09:10:00 | 2026-09-30T09:40:00

== aliases (one row per alias, unique) ==
entity | alias
Campus Navigator | Campus Navigator
Campus Navigator | navigator
Campus Navigator | the app
DBMS project team | DBMS project team
DBMS project team | project team
DBMS project team | the team

```

Figure 10: The sqlite3 section of the same run. Three facts are current. At 09:30 the owner was the DBMS project team. Aliases are one row each.

7. Testing and observations

The demo ends in nine checks, all passed on the run in the figures: alias resolution, a single active owner, the 09:30 as-of result, the forgotten MongoDB fact absent from `current_belief`, a SQLite hit for "full-text search sqlite", an empty result for "MongoDB", a new fact id from supersede, the summary naming AI Lab, and `PRAGMA integrity_check`. The process exits 0 only if every check passes. Two consecutive runs printed the same transcript.

Two results are easy to mix up. First, lexical retrieval of "who owns Campus Navigator" ranks episode 2 (the DBMS project team) above fact 5 (AI Lab). The old sentence is still in the episode table, and it shares the query's words. The question "who owns it now" is the predicate query on `current_belief`, which returns AI Lab only. The as-of query is what returns the earlier owner. Keyword search and the validity interval answer different questions, and the transcript shows both. Second, forgetting removes the MongoDB rows from retrieval without deleting the entity. The entity and its alias remain, which is what a registry is for.

The foreign-key refusal in Figure 9 happens on the Python connection, which has executed `PRAGMA foreign_keys = ON`. A bare `sqlite3` shell does not turn that pragma on by itself. The keys are declared either way. Enforcement is per connection. `queries.sql` still prints `PRAGMA foreign_key_list(fact)`, and that listing has the four foreign keys on `fact`.

8. Limitations and future work

The feature hash measures token overlap. It is not a semantic embedding, and similarity is a scan of a few dozen vectors rather than an approximate index. A model can replace embed without a schema change: the column is already a vector of floats. Consolidation does not implement Nemori's predict-calibrate loop or LightMem's sleep-time update. It stores the active facts of one session. The partial unique index is weaker than the PostgreSQL exclusion constraint in Phase 2: it limits active rows, and it does not reject two closed intervals that overlap. The demo is a single writer, so it does not reproduce a lost update. The unique index is what would make that race fail instead of silently keeping both rows. There is no authentication.

Harness the Memory found that no one substrate wins every task. This program already runs a predicate query beside the fused lexical lanes. A later version can route a question to one or the other instead of always printing both. Moving the same schema to PostgreSQL would be the place to add the range exclusion constraint SQLite does not have.

9. Conclusion

TA-3 is a runnable SQLite memory for one agent. Episodes, entities, aliases, facts, keywords, embeddings, and a session summary are ordinary tables with keys and checks. The session writes six turns, refuses a second current owner until the old interval is closed, forgets a speculative fact without losing the audit trail, and answers "who owns Campus Navigator" both as a predicate query (AI Lab, from 09:40) and as a fused retrieval over the text. Nine checks pass from a clean database file.

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